

TOLGAHAN AY

TECHNICAL GAME DESIGNER & GAMEPLAY PROGRAMMER INTERN

Muğla / Bodrum, Türkiye | +90 534 273 43 00 | tolgahan.ay@outlook.com

INTERACTIVE PORTFOLIO & CASE STUDIES
https://togi1.github.io
github.com/Togi1

Multidisciplinary Game Designer & Gameplay Developer combining C# systems architecture, data-driven game balance, and visual asset integration with an engineering background holding **two registered hardware patents**. Experienced in building playable 2D/3D Unity prototypes, designing core loops and rule engines, and coordinating team pipelines via Git, Jira, and Steamworks.

MULTIDISCIPLINARY SKILL MATRIX

Systems Design • Gameplay Code • Production

Game & Systems Design: Core Loops, Rule & Dilemma Systems, Dynamic Economy & Progression Balancing, GDD Authoring, Game Feel

Unity Engine Systems: ScriptableObject Data Architectures, 2D/3D Physics (`Rigidbody`), Weighted RNG Loot Tables, Custom UI

Programming & Architecture: C# (OOP, Event-Driven Systems, Coroutines), Python (Data Structures & Algorithms), State Management

Pipeline, Art & Hardware: Git/GitHub, Unity Cloud, Jira, Steamworks Packaging, Blender (3D Asset Integration), 2D Pixel Art, Robotics

SELECTED GAME PROJECTS & TECHNICAL CASE STUDIES

Live demos & source code at togi1.github.io

Project PM535-T — Game Designer & Project Manager

`2D Point-and-Click Simulation | In Development`

- Authored the comprehensive Game Design Document (GDD) and led a 4-person multidisciplinary team across a structured 5-phase production roadmap.
- Designed a dystopian border-quarantine diagnostic rule engine inspired by *Papers, Please*, balancing misleading clinical symptoms, strict test-quota constraints (2/3 limits), and moral dilemma mechanics.

Pixel Angler — Solo Gameplay Programmer, Economy Designer & 2D Artist

`Unity 2D, C# | Rapid Prototype (3 Days)`

- Architected a designer-friendly, data-driven fish pool using `FishData` ScriptableObjects and a cumulative weighted-random spawn algorithm (`spawnChance`) to control rarity tiers without code edits.
- Coupled catch difficulty directly to mini-game movement speed and dynamic market pricing (`GetDynamicPrice()`), paired with an exponential shop upgrade economy (×1.5 cost scaling) and original 2D pixel art.

3D Mobile Dart Prototype — Solo Gameplay Programmer

`Unity 3D, C# | Rapid Prototype (3 Days)`

- Translated drag-and-release touch/mouse input vectors into 3D impulse physics (`ForceMode.Impulse`), dynamically aligning projectile orientation with real-time velocity vectors every frame for believable flight trajectories.
- Engineered Inspector-tunable collision and bounce resolution alongside multi-layered player feedback (camera follow, audio cues, haptics, and impact particle prefabs); planned refactoring of `Invoke` and object lookups into C# Events and Coroutines.

BlackFriday (Team Project) — Release Pipeline & Technical Coordination

`Unity Cloud, Jira, Steamworks`

- Coordinated multi-developer version-control workflows via Unity Cloud and tracked sprint tasks in Jira, while executing build packaging and deployment configurations for Steamworks.

REGISTERED HARDWARE PATENTS

- **Autonomous Line-Tracking Robotics Platform**
  `Patent Reg: TR2022017364U5` — Embedded sensor integration & autonomous navigation logic.
- **Multi-Functional Storage & Peripheral Hardware Unit**
  `Patent Reg: TR2022018474A2` — Modular hardware ergonomics & peripheral architecture.

Languages: Turkish (Native), English (B2 Professional Working) | **Interests:** Systems Design, Rapid Prototyping, Licensed Badminton Athlete

EDUCATION & HONORS

Muğla Sıtkı Koçman University (2025 – Present)
BSc in Digital Game Design • *High School Valedictorian (1st Rank)*

- **Foundational C# Certification** — Microsoft
- **Robotic Coding Specialist** — Advanced Embedded Systems
- **DiscoverEU Laureate** — European Commission Mobility